yoinked from https://github.com/Nicicalu/OST-CN2-Spick

## OSPF (IGP)

ECMP load balancing, fast convergence, widely used. IP Port 89 on L3 (no TCP, has own header). Uses `224.0.0.5` (all routers), `224.0.0.6` (all DR,BDR).
**DR Election**: Highest IF prio → if tie, highest Router ID. Priority 0 = ineligible for DR/BDR. No preemption. BDR = same rules, second best candidate
**Virtual links**: Cannot go through more than one area, can only run through standard non-backbone areas. (eg. not over stubby areas)

### ROUTER OPERATION

– Establish neighbor adjacencies and exchange LSAs
– Build the Link-State Database (LSDB)
– Run Dijkstra's SPF algorithm:
  **Intra-area change**: SPF recalculation
  **Inter-area change**: No SPF needed – ABR handles updates
– Build the routing table from SPF results

### ROUTE TYPES

**O**: Intra-area: Local area → Local area. Type 1 and 2 LSAs
**O IA**: Inter-area: Local area → Neighboring area. Via LSA 3 through backbone
**O E1 or O E2**: External: Local area → External (eg. BGP), from ASBR
**O E1 / O N1**: Cost = internal + external metrics (**def=20**)
**O E2 / O N2**: Cost = external metrics. Default external route for OSPF
**Cost calculation**: Reference Bandwidth(def=100 Mbps)/Interface Bandwidth

### AREAS

AS split into **areas** with 32-bit Area ID (e.g., `0.0.0.0` = Area 0)
**Backbone Area (Area 0)**: Core of the OSPF domain, must connect to all other areas (directly/virtual links), must be contiguous (no disjointed segments), should not contain end-user networks, summarizes topology of other areas
**Non-Backbone Areas**: Connect end users and local resources, All inter-area traffic must transit the backbone

### ROUTER TYPES

**Area Border Router (ABR)**: Connects two or more OSPF areas, must have 1 interface in backbone. 1 OSPF DB per Area
**Internal Router (IR)**: All interfaces belong to the same area (non-backbone)
**Backbone Router (BR)**: At least one interface in Area 0, Includes ABRs and routers internal to the backbone
**AS Boundary Router (ASBR)**: Connects to external AS/Network (e.g., BGP), Advertises external routes into OSPF, can be in backbone or non-backbone

### DESIGN RULES

**Rule 1**: Backbone (Area 0) must be contiguous – no partitions allowed
**Rule 2**: Every non-backbone area must connect to Area 0

### LINK STATE ADVERTISEMENT (LSA) TYPES

**(1) Router-LSA**: ***All routers*** list directly connected links (so all outgoing interfaces with state and cost) (**intra-area**)
**(2) Network-LSA**: ***DR*** lists all routers and DR in broadcast/multi-access network (**intra-area**)
**(3) Summary-LSA**: ***ABR*** advertises networks from other areas, flooded in all the areas that are not "totally stubby" (**inter-area**)
**(4) ASBR-Summary**: ***ABR*** advertises path to ASBRs. ASBR sends Type 1 + external bit, ABR converts to Type 4 and floods (**inter-area**)
**(5) AS-External-LSA**: ***ASBR/ABR*** advertises external routes (e.g., BGP); flooded to all non-stub areas → only normal areas (**inter-area**)
**(7) NSSA External**: Like Type 5 but inside NSSA. Converted to Type 7 by ABR. Shown in routing table as **(O N1 or O N2)**

### AREA TYPES (ALLOWED LSA TYPES)

Stubby = Blocks external LSA [No 5], Totally = Blocks inter-area LSA [No 3/4]
Default route (`0.0.0.0`) from ABR [DR ABR], Cannot contain ASBR [No ASBR]
**Standard (1,2,3,4,5)**: Normal
**Stub Area (1,2,3)**: [No 4/5] [DR ABR] [No ASBR]
**Totally Stubby Area (1,2)**: [No 3/4/5] [DR ABR] [No ASBR]
**Not-So-Stubby Area (1,2,3,7)**: [No 4/5], no DR, allows 1 ASBR [LSA 7 → 5]
**Totally NSSA (1,2,7)**: [No 3/4/5] [DR ABR] [LSA 7 → 5]

### PACKET TYPES

**(1) Hello**: Usually Multicast. Discover, maintain, and verify neighbors. Forms adjacencies. Election of DR,BDR in broadcast networks. Contains network mask (sending routers' IF), Hello interval (p2p,broadcast: def=10s), Options, Priority (election), Router dead interval (def=40s), DR/BDR IP, Neighbors.
**(2) Database Description (DBD/DD)**: Unicast. Exchange summaries of LSAs during adjacency formation (headers only). Contains IF Max. MTU, Options, I/M/MS bits (Initial, More, Master-slave bit), DD Seq. Num., LSA Header
**(3) Link State Request (LSR)**: Unicast. Request specific LSAs. Contains Link State Type (router/net), Link State ID, Advertising Router (sender address)
**(4) Link State Update (LSU)**: Multi/Unicast. **Implicit ack.** Flood new/updated LSAs, Answer to LSR (unicast), Contains Nr. of LSAs, full LSAs information.
**(5) Link State Acknowledgment (LSAck)**: Multi/Unicast. **Explicit ack.** Confirms receipt of LSAs for reliable flooding. Acks may be grouped together.

### SUB-PROTOCOLS

**Hello Protocol**: Neighbor discovery. Parameter negotiation. Maintain logical adjacencies on P2P, P2MP, virtual links. Elect DR/BDR on broadcast and NBMA networks. Continuously sends hello packets to maintain bidirectional connectivity; failure to receive = neighbor down (RouterDeadInterval)
**Database Sync Protocol**: Sync LSDB using DBD packets.

#### NEIGHBOR STATES

**Down**: Initial state, no Hello packets exchanged or after RouterDeadInterval
**Init**: Received Hello packet from neighbor but without its own Router ID yet
**2-Way**: Seen its own Router ID. DR/BRD election is taking place in NBMA
**ExStart**: (DBD) Bi-dir comm; highest Router-ID=master. Determine init Seq Nr
**Exchange**: Exchange of DBD packets (LSA headers).
**Loading**: Missing LSAs are requested. (LSR/LSU)
**Full**: Databases fully synchronized. Normal operating state

### OSPF ROUTING AND ECMP

– Each router runs Dijkstra per area; link cost = metric from LSAs (1–65535)
– Routes added to RIB/FIB based on computed next hops
– ECMP: Modified Dijkstra supports Equal-Cost MultiPath if multiple paths have same cost → routes added with multiple next-hops for load balancing

### PATH SELECTION

Path always has to go through backbone
1) Most specific match (CIDR)
2) O > O IA > E1 > N1 > E2 > N2
3) Lowest link cost (reference bandwidth/interface bandwidth)

### POSSIBLE FAILURES

– No OSPF enabled / Interface not in OSPF / Duplicate router ID
– Passive interface that has to be removed / Virtual Link badly configured
– Area ID or Area Type mismatch on neighbors
– Hello interval or Router dead interval mismatch

## IS-IS (IGP)

Widely used by ISPs, Scalable, Fast failure detection & convergence, ECMP, Extendable, Less CPU intensive, Default metric of 10 on most routers
**ES**: End-host devices are called End Systems (ES)
**IS**: Routers are called Intermediate Systems (IS)

### CONNECTIONLESS NETWORK SERVICES (CLNS)

**CLNS**: ISO Layer 3 datagram service; supports **CLNP**, **ES-IS**, **IS-IS**.
**CLNP**: Connectionless Network Protocol, similar to IP, used in ISO stack
**ES-IS**: host ↔ router discovery protocol
**IS-IS**: router ↔ router. Link-state routing protocol
**Integrated IS-IS**: Allows IP routing with IS-IS

### NETWORK SERVICE ACCESS POINT (NSAP)

Variable in length (up to 20 bytes), Identifies node, not IF. Example from IP: `192.168.1.24` → `49.0001.1921.6800.1024.00`

```
49.0011.0000.0000.0003.00
   Area ID (11)  System ID (R3)  NSEL
```

**Network Entity Title (NET)**: Unique router identifier. NSEL=0,Starts with 49
**Authority & Format Identifier (AFI)**: NSAP format+authority that assigned it
**Initial Domain Identifier (IDI)**: Variable len, identifies administrative domain
**Domain Specific Part Format Identifier (DFI)**: Specifies format
**Domain Specific Part (DSP)**: Var len, contains hierarchical structure of addr
**NSEL (N-Selector)**: always `00` for routers (1b)
**System ID**: Unique per router (often based on lo-IP/MAC) (6b)
**Area ID**: Used for routing hierarchy (like OSPF areas) (AFI+IDI+DSP)

### PACKET TYPES (ALL IN L1 OR L2)

All PDUs contain Header with shared common fields + specific routing-related information (Type, Length and Value (TLV)) used to extend protocol
**IIH (Hello)**: Build,maintain adjacencies. Includes system ID, holding time, prio
  **Functions**: discover, build, maintain
  **Interval**: 10s (default); DIS sends every 3.3s on LANs
  **Multiplier**: Missed Hello limit → Holdtime = Interval × Multiplier(def=3)
  **Multicast**: [L1 `01-80-C2-00-00-14`] [L2 `01-80-C2-00-00-15`]
**LSP (Link State PDU)**: Contains topology info including prefixes with costs; unicast on p2p or flooded throughout the area (OSPF: LSA Type 1)
**CSNP (Complete SNP)**: Sent by DIS; lists all known LSPs (OSPF: DBD)
**PSNP (Partial SNP)**: Used to request missing or acknowledge received LSPs

### ROUTER TYPES

**L1 Router**: Within one area; all have same LSPDB. No inter-area routing
**L2 Router**: Backbone router; all have same LSPDB. Routes between areas
**L1/L2 Router**: Acts as both; separates LSPDBs, redistributes between levels

### ADJACENCIES

Backbone must be contiguous: (L2 connection) L1 ↔ L1-L2 ↔ L2. NOTE: L1/L2 routers form both L1 **and** L2 adjacencies if they are both in the same area. Interface MTU and auth key must be identical to become neighbors
**L1**: Area addresses match unless configured otherwise
**L2**: Alongside L1, unless router is L1 only, areas must not match

#### POINT-TO-POINT (NO DIS)

Initialized by receipt of ISHs, exchange p2p IIHs padded to MTU
**CSNP**: Sent once at adjacency startup
**LSP**: Advertises topology changes (link-state info)
**PSNP**: Acknowledges received LSPs or requests missing ones

#### BROADCAST/MULTI-ACCESS (DIS REQUIRED)

Not triggered by ISHs, instead broadcast IIHs with all neighbors MAC's
**Designated Intermediary System (DIS)**: Create, update pseudonode LSPs. Flood LSPs. Similar to DR in OSPF. No backup DIS.
**DIS Election**: Highest priority (0–127) (Cisco def=64) > Highest SNPA (MAC)
**Preemption**: Enabled → higher prio router automatically takes over DIS Role
**Pseudonode**: Carried out by DIS. Multiaccess links. Separate for L1,L2. Fake node IS-IS creates to represent a broadcast LAN
**Passive interfaces**: Advertise network prefixes without adjacency forming

### PATH SELECTION

Path always goes through first BR if no route leaking
Default interface metric (m) = 10 (**Lower Metric better**)
1) Most specific match (CIDR)
2) L1 intra-area > L2 intra-area > Leaked L2 → L1 (internal m) > L1 external (external m) > L2 external (external m) > Leaked L2 → L1 (external m)
3) Lowest cost
4) Lowest system ID
5) Lowest interface

### ROUTING

Summarization occurs when routers enter an IS-IS level (L1 → L2, L2 → L1).

#### LEVEL 1

***Intra-area*** routing only (L1 Area like OSPF Totally Stubby Area)
**L1**: routers install a **default route** to nearest L1-L2 for inter-area traffic
**L1-L2**: Do **not** advertise L2 routes into L1 area (unless route leaking active)
  Set **Attached bit** to signal L2 connectivity to backbone
**Distribution Bit**: Set to 1 on L2 → L1; blocks re-advertisement L1 → L2.
**Route-Leaking**: injects a more specific route into L1 to improve routing

#### LEVEL 2

Routing between areas (***inter-area***). L1-L2 routers inject L1 routes into L2 topology. L1 routes are redistributed into L2 with L1 metric preserved in L2 LSP

### IS-IS VS OSPF

| *Feature* | *IS-IS* | *OSPF* |
|---|---|---|
| Layer | L2 (CLNS) | L3 (IP, Port 89) |
| Encapsulation | No IP, uses TLVs | IP packets |
| Hello Type | IIH | Hello packet |
| Area Model | L1/L2 | Backbone + Areas |
| Metric | Cost (default 10) | Cost (bandwidth) |
| Router ID | System ID (6B) | 32-bit Router ID |
| Adj. Types | L1, L2, L1/L2 | DR/BDR, P2P |
| LSDB | Per level (L1/L2) | Per area |
| Scaling | Large-scale ISP core | Enterprise/campus |
| Routing Info | TLVs (flexible) | Fixed LSA types |
| Type | Link-State | Link-State |
| Shortest Path calculation | Dijkstra | Dijkstra |

### POSSIBLE FAILURES

– Wrong addressing (NET, Sys-ID, Area ID) / Interface type (P2P, broadcast)
– Wrongly configured L1,L1-L2,L2 routers (eg. no L1-L2 as transit)
– Authentication mismatch (TLV auth → configure identical IS-IS auth)

## BGP (EGP)

Path-vector routing protocol (doesn't contain full topology). Stable, flexible, scalable. Based on TCP. Loop prevention using AS_PATH

### IBGP

Routers in same AS, AD=200, more trusted (lower security overhead). AS-Path and Next-hop not modified

#### SCALABILITY

– iBGP does not re-advertise routes between iBGP peers → full mesh required (cause no loop prevention)
– Session count = $^{n(n-1)}/_{2}$ (e.g., 5 routers = 10 sessions, 10 = 45)

#### ROUTE REFLECTORS (RR)

Solves iBGP full-mesh (***split horizon***) scaling by selective route refl. Clients only peer with RR; unaware they're clients. Only RR needs special config
**Clients**: iBGP neighbors that rely on the RR for route sharing
**Non-clients**: regular iBGP peers; still have full-mesh among themselves
**ORIGINATOR_ID**: stops a router from accepting its own reflected route
**From non-client**: RR advertises to **clients only**
**From client**: RR advertises to **all** (clients + non-clients)
**From eBGP peer**: RR advertises to **all** (clients + non-clients)

### EBGP

Routers in different ASes, AD=20, stricter policy enforcement. Prepends ASN to AS-Path, modifies next-hop IP. Fails if own ASN in AS-Path. TTL=1

**AS-Override**: Allows reuse of same ASN across different customer sites (e.g., Swisscom); rewrites ASN to avoid loop detection
**next-hop-self**: Router can't install a route if the next hop address is unreachable ⇒ next-hop-self changes next hop address to own local addr by BR

### AUTONOMOUS SYSTEM NUMBERS (ASN)

– Unique ID for each AS; required for Internet routing with BGP
– Private: [Legacy 16-bit 64'512-65'535] [32-bit 4'200'000'000-4'294'967'294]

### PEERING / NEIGHBORS

– Two routers with a BGP TCP session (port 179) are called peers/neighbors
– Each BGP router is a **BGP speaker**
– Supports CIDR, route aggregation; decisions based on policies/rules

### PATH ATTRIBUTES

Used for route control and policy enforcement
**Well-known mandatory**: Always present (e.g., AS-Path, Origin, Next Hop)
**Well-known discretionary**: Optional but recognized by all (e.g., Local Pref)
**Optional transitive**: Passed between ASes (e.g., Community, Aggregator)
**Optional non-transitive**: Not passed across ASes (e.g., MED, Weight)
**NLRI**: Routing table info: prefix, prefix length, and associated path attributes

### MESSAGES

**OPEN**: Establishes session; includes version, ASN, Hold Time, BGP Identifier, optional params. **Hold Time:** Heartbeat in seconds (Cisco def=180s), reset by KEEPALIVE/UPDATE; 0 = session down. **BGP Identifier:** 32-bit Router-ID, manually set or highest loopback/active IP; used for loop prevention
**KEEPALIVE**: Sent every 1/3 of Hold Time; ensures neighbor liveness
**UPDATE**: Advertises new routes, withdraws old ones, or both; includes NLRI (prefix + path attributes); can act as KEEPALIVE
**NOTIFICATION**: Sent on session error (e.g. hold timer expired); terminates session immediately

### NETWORK STATEMENTS

Advertise a specific network prefix to BGP peers (only the best is advertised, even if multiple exist). Must exist in the RIB so that it's imported in BGP table
**Connected**: next-hop: `0.0.0.0`, origin attribute: `i`, weight: 32'768 (Cisco)
**Static/RP**: next-hop: NH in RIB, MED: IGP metric, rest same as above

### BEST PATH CALCULATION

– BGP maintains all received paths per prefix but advertises only the best one
– Best path is installed in RIB; recalculated on:
  – Next-hop reachability change
  – Interface failure to eBGP peer
  – Redistribution change
  – New/withdrawn path received
– **Influence:**
  – Outbound BGP policy → inbound traffic behavior
  – Inbound BGP policy → outbound traffic behavior

#### PATH SELECTION (IF PREFIXES ARE THE SAME):

1) Prefer highest **Weight** (**Cisco-specific**, local to router)
2) Prefer highest **Local Preference** (global within AS)
3) Prefer routes **originated by the router** (only small i in path, NH `0.0.0.0`)
4) Prefer **shorter AS path** (only length is compared)
5) Prefer **lowest origin type:** IGP < EGP < Incomplete (I on origin)
6) Prefer **lowest MED** (Multi-Exit Discriminator) (also called metric)
7) Prefer **external (EBGP)** over **internal (IBGP)**
8) For iBGP: prefer path with **lowest IGP metric** to next-hop
9) For eBGP: Prefer **oldest** (more stable) path
10) Prefer **lowest BGP router ID**
11) Prefer path from **lowest neighbor IP address**

### ROUTE FILTERING

– Filters control which routes are received/advertised
– Used for security, traffic shaping, memory optimization
– Tools: **prefix-list** (IP), **filter-list** (AS-path), **route-map** (flexible match/set)

### COMMUNITIES

32b optional, transitive tag (e.g. `ASN:value`, `65000:100`). Used to mark routes for policy control across ASes. Can be added, modified, or removed at each hop

### ENTERPRISE CONNECTIVITY OPTIONS

**Single/Dual**: denotes how many **links** there are
**Multi-Homed/-Homed**: denotes how many **ISPs** are connected
**Single-Homed**: One ISP, one link (BGP or static). Simple but no redundancy
**Dual-Homed**: One ISP, two links (routers). Redundancy within same provider
**Multihomed**: Multiple ISPs. improved redundancy and routing control, but avoid being a transit – advertise only customer-owned prefixes
**Dual-Multihomed**: Multihomed, but two links per ISP. Most redundancy

#### TRAFFIC ENGINEERING (TE) - INBOUND VS OUTBOUND

**Local Pref**: [OUT] [highest ↑] Lower local pref on backup path
**Multi-Exit Discriminator (MED)**: [IN] [lowest ↓] Higher MED on backup path
**AS-Path Prepending**: [IN] [shortest ←] Add multiple own ASN on backup path
**TE Limitation**: AS controls [OUT], [IN] control limited - ISPs may ignore
**TE with Aggregate**: Prefer primary ISP with summarized routes; advertise specific prefixes on backup for failover
**Transit**: One ISP provides reachability to all destinations
**Peering**: ISPs provide each other reachability to portions of their routing table

### TRANSIT VS. PEERING

**Transit**: ISP provides full reachability (paid relationship)
**Peering**: ISPs exchange selected routes; equal relationship, usually unpaid
**AS Path Filtering**: to avoid getting transit: ip as-path access-list 10 permit ^ $

### INTERNET EXCHANGE POINT (IXP)

Facility where networks exchange traffic via BGP peering, ***routing policies*** determine which routes are advertised/accepted. Reduces transit costs, latency, and offloads upstream links

#### PUBLIC PEERING

Members peer via shared switch fabric and a ***route server***, which distributes routes but stays out of data path (NEXT_HOP unchanged). Simplified setup (one legal contract), minimal policy control; one BGP session to route server

#### PRIVATE PEERING

Direct BGP sessions between two parties (***1:1***), requires one session and legal contract per peer. May use **public or private** interconnects. Full policy control per neighbor. **Examples:** Equinix, SwissIX (non-profit)

### RESOURCE PUBLIC KEY INFRASTRUCTURE (RPKI)

– Framework to validate that prefix is legitimately originated by a specific ASN
– Prevents route **hijacking** and accidental mis-originations (route **leaks**)
– RPKI validates origin only – not the AS-path

#### KEY COMPONENTS

**Trust Anchors (TAs)**: Root CAs of the 5 RIRs; issue certs for resource holders
**Route Origin Authorization (ROA)**: Digitally signed object, authorizes ASN to announce prefix (AS, prefix that AS can originate, max prefix length)
**RPKI Validators**: downloads+verifies+stores Validated ROA Payloads (VRPs)

#### RPKI VALIDATION STATES

**Valid**: Prefix + ASN match ROA
**Invalid**: Prefix found, but ASN mismatch or prefix too long
**Not Found / Unknown**: No matching ROA

#### VALIDATOR DETAILS

– Use TALs (Trust Anchor Locators) to fetch data from RIRs
– Validate cryptographic signatures (via X.509 certs with RFC 3779)
– Outputs VRPs; invalid objects are discarded
– Update frequency: >=24h (recommended), ~30–60min (practice)
– RRDP (HTTPS) to fetch RPKI repository data (RFC 8182) replacing rsync
– RPKI-RTR: used to get the validator's already-processed cache of VRPs

### MONITORING

Event tracking, BGP hijack detection, Route leak detection, RPKI status check, Reachability tracking, AS path change tracking, AS path visualization

## MULTICAST

**Use-cases 1-to-Many**: Streaming, software updates, music-on-hold
**Use-cases Many-to-Many**: Gaming, VR, stock data, group chat
**Benefits**: Efficient bandwidth, lower server/CPU load, no redundancy
**Properties**: UDP-based. Apps must handle drops, duplicates, out-of-order
**Source**: Sends to group IP; doesn't need to join
**Receiver**: Must explicitly join group to receive traffic

### MULTICAST ADDRESS RANGES

`224.0.0.0 – 224.0.0.255`: Link-local, TTL=1 (not forwarded by routers) (IGMP)
`224.0.1.0 – 224.0.1.255`: Reserved by IANA, routable (NTP)
`232.0.0.0 – 232.255.255.255`: Source-Specific Multicast (SSM)
`235.0.0.0 – 238.255.255.255`: Globally scoped multicast (routable)
`239.0.0.0 – 239.255.255.255`: Administratively scoped (private space)

### BROADCAST BASICS

L3 routes between subnets; L2 floods within same subnet
**L2 (Bridging)**: MAC `ffff.ffff.ffff` switches flood to all ports in VLAN
**L3 (Routing)**: `255.255.255.255`: local broadcast, not routed. Directed broadcast for specific subnet (e.g. `10.1.1.255`): can be routed if enabled

### L2 MULTICAST: MAC MAPPING

1) **Get IP:** Example multicast IP address: `239.5.5.5`
2) **Convert to Binary:** `239.5.5.5` = 11101111.00000101.00000101.00000101
   → Take only the last 23 bits: 0000101.00000101.00000101
3) **Map to MAC Prefix:** Fixed `0100.5E` → Map last 23 bits to: `0100.5E05.0505`

### INTERNET GROUP MANAGEMENT PROTOCOL (IGMP)

**Purpose**: Manages group membership for IPv4 multicast on each segment
**IGMPv1**: Basic join via query-response mechanism, no explicit group leave. Router sends ***membership queries*** every 60s to `224.0.0.1`, removes group after timeout if no report received. Receiver doesn't know multicast source
**IGMPv2**: Adds ***Leave Group*** message. Supports ***Group-Specific Queries*** e.g. when someone leaves group ('anyone still interested in group xy?'), reducing broadcast overhead. Receiver still doesn't know who the source is
**IGMPv3**: Adds ***source filtering*** (Include/Exclude lists). Enables ***Source-Specific Multicast (SSM)*** – receiver requests traffic only from selected source(s), **no need for Rendezvous Point** (RP) anymore. Adds support for application-level access control and filtering. Can also be used in ASM (Any Source Multicast), but mainly with SSM. (`224.0.0.22` all IGMPv3 routers)

#### IGMP SNOOPING

**Without snooping**: multicast = broadcast on VLAN
**With snooping**: switch listens to IGMP messages & builds a forwarding table
**Default**: snooping is enabled; switch needs IGMP Querier to operate

### REVERSE PATH FORWARDING (RPF)

To avoid loops, verifies that a multicast packet arrives on the interface that a unicast packet destined for the multicast source would be forwarded out of.

### RENDEZVOUS POINT (RP)

Used in Shared Tree (`*,G`) setups with PIM-SM, acts as common meeting point. **RPF check** is performed toward the **RP** (not the source). Once **source is known**, routers may **switch** to a Source Tree (`S,G`). Triggered by SPT join timer expiration/RPF check confirming better route/data rate exceeding

### SHARED TREE VS. SOURCE-BASED TREE

**Source-Based Tree (`S,G`)**: Built and added to ***mroute table*** when router receives an (`S,G`) join/report from IGMP host. `S` = known source; `G` = group
**Shared Tree (`*,G`)**: IGMP host sends ***membership report*** (IGMP Join), Router adds (`*,G`) entry to ***multicast routing table*** `*` = any/unknown

### PROTOCOL INDEPENDENT MULTICAST (PIM)

Only between routers. Relies on unicast routing table (RIB) for multicast forwarding decisions and RPF. Protocol-independent.

### PIM DENSE MODE (PIM-DM) – NO RP

**Push Model**: Floods multicast to all interfaces; prunes where no receivers
**(1) Flooding**: Source sends traffic to multicast-enabled links using unicast RIB
**(2) Distribution Tree**: Initially entire network (shared tree rooted at source)
**(3) Prune Messages**: Routers without interested receivers send prunes upstream to remove themselves from the tree
**(4) State Maintenance**: Routers track source, receivers, interfaces to forward/prune per group

### PIM SPARSE MODE (PIM-SM) – ASM, SSM

**Pull Model**: Multicast traffic only sent where requested (***explicit join***). Works with IGMP to detect interested receivers, uses unicast routing for forwarding
**(1) Join/Prune**: Routers send explicit join/prune messages to request or stop receiving multicast for group (`G`) to other routers
**(2) Forwarding**: Routers only forward multicast packets for group (`G`) on interfaces from which explicit joins were received

#### ANY SOURCE MULTICAST (ASM)

– Works with **IGMPv1** or **IGMPv2** → receiver does not know the source
– Receiver sends ***IGMP Join (`*,G`)*** to its first-hop router
– First-hop router forwards ***PIM Join (`*,G`)*** hop-by-hop toward the RP
– Sources send multicast traffic to the RP via a ***PIM Register*** tunnel
– All routers in the multicast domain must know the RP location

#### SOURCE SPECIFIC MULTICAST (SSM)

– Receiver subscribes using **IGMPv3**, providing both source (`S`) and group (`G`) to the first-hop router (**explicit join**)
– No RP required → **PIM-SSM builds only (`S,G`)** Shortest Path Trees (SPT)
– **No shared tree** (`*,G`) model used; SSM is source-directed
– IANA reserved `232.0.0.0/8` for SSM in IPv4
– Join msgs forwarded hop-by-hop toward source to establish forwarding path
– Uses unicast routing table (RPF) to maintain loop-free delivery

#### PIM SPARSE-DENSE MODE

**PIM Sparse Mode**: Pull model – multicast traffic is forwarded only on request
**PIM Dense Mode**: Push model – traffic is flooded everywhere, then pruned
**Sparse-Dense**: Supports both per multicast group, depends on RP availability

#### PIM DENSE MODE VS. PIM SPARSE MODE USE-CASES

**DM**: Small/tightly scoped networks where most devices need multicast
**SM**: Large-scale/distributed env where multicast receivers are few/spread out

## NETWORK DESIGN

**Pillars:** Scalability, Speed, Availability, Security, Manageability → overall cost

### AVAILABILITY

**MTTD**: Mean Time To Detect
**MTTI**: Mean Time To Identify
**MTTR**: Mean Time To Repair
**MTTRS**: MTT Restore Service
**MTBF**: Mean Time Between Failures
**MTTBSI**: MTB Service Incidents
**MTBF combined**: $\left(\sum_{n=1} {}^{1}/_{MTBF_n}\right)^{-1}$
**MTBF parallel**: $\sum_{n=1} {}^{MTBF_n}/_{n}$
**Availability**: $^{MTBF}/_{MTBF+MTTR}$
**Better Av**: MTTR → 0 ∧ MTBF → ∞

### REDUNDANCY

Adds reliability, increases MTBF but increases MTTR (complexity)
**Backup Paths**: Duplicate devices/links on primary path, build extra links for redundancy, consider backup link capacity, consider failover speed
**Load Balancing**: ECMP, EtherChannel, Port-Channel

### TOPOLOGY

Star (flexible → office), Ring (cheap → city), Bus, Tree, Full Mesh

#### HIERARCHICAL DESIGN

**Core**: L3, Highspeed backbone, no policy, scalable/redundant, no security
**Distribution**: L3, policy control, HSRP/VRRP, loop protect., small fault domain
**Access**: L2, Connect end devices, high port count, port security, QoS marking
**Switch stacking**: Merging physical switches into one large logical switch
**Collapsed Core**: Combines Core + Distribution (small/medium networks)

#### FLAT DESIGN

Any-to-any; small networks, MPLS/LAN setups, lacks scalability and control

#### FABRIC DESIGN (EMERGING TECHNOLOGIES)

Modern design using ***Underlay*** (transport, e.g. IP, MPLS) / ***Overlay*** (logical virtual topology, e.g. EVPN) with ***Software-Defined Networking*** (SDN) and ***Software-Defined Access*** (SDA)

### ENTERPRISE CAMPUS

– 1000s of users, multiple buildings, one physical location, one company
– Multiple interconnected LANs, connected via Ethernet and Wireless
– Flat design for client access, Hierarchical design for the rest (per building)
– EVPN, MPLS are used to reduce size of L2 domain

### FIRST HOP REDUNDANCY PROTOCOLS (FHRP)

Ensures default gateway is always reachable, PCs can only have one. Enables fast failover during router failure
**Virtual Router Redundancy Protocol (VRRP)**: Shared virtual IP, real MACs per router, Master router handles forwarding, rest as backup
**Hot Standby Router Protocol (HSRP)**: (Cisco) Shared virtual IP + virtual MAC, One active, others in "hot standby" → faster
**Gateway Load Balancing Protocol (GLBP)**: (Cisco) HSRP + Load balancing + redundancy. **Active Virtual Gateway (AVG):** Answers ARP and sends virtual MACs of AVFs. **Active Virtual Forwarder (AVF):** Forwards traffic

### DATA CENTER

| *Parameters* | *Tier 1* | *Tier 2* | *Tier 3* | *Tier 4* |
|---|---|---|---|---|
| *Uptime guarantee* | 99.671% | 99.741% | 99.982% | 99.995% |
| *Downtime per year* | <28.8 hours | <22 hours | <1.6 hours | <26.3 minutes |
| *Price* | $ | $$ | $$$ | $$$$ |
| *Compartmentalization* | No | No | No | Yes |

#### DESIGNS

**Top of Rack (ToR)**: switches per rack. less cabling, easy expansions/exchanges per 'rack', scalable glass fiber, ideal for high service density. more switches, more ports, more L2 Srv-2-Srv traffic, more STP
**End of Row (EoR)**: switches per row. less switches, higher utilization of ports, switches all at one place, better L2 availability between racks. more cabling

#### TRAFFIC DIRECTIONS

**North–South**: Between external networks (client-server, in/out DC)
**East–West**: Within DC (e.g., server-server, storage)

#### THREE-TIER ARCHITECTURE

– Access → Aggregation → Core (like Hierarchical)
– Optimized for North-South traffic, Not ideal for East-West (VMs, containers)

#### LEAF-SPINE ARCHITECTURE

– Two-tier: Leaf switches (access) connect to Spines (core)
– High performance, low latency, Scalable, ideal for East–West traffic

## WAN

Connects remote LANs via SPs for data/voice/video; key needs: bandwidth, control, design, resilience, mgmt. **Requirements:**
**Bandwidth**: App needs, peak usage, reserve for VoIP
**Control/Security**: Trust provider? No full control
**Availability**: Redundancy, SLA for failures
**Mgmt**: Inband vs out-of-band

### PRIVATE WAN

**Point-to-Point**: Leased L2 line (Ethernet); monthly fee; private circuit; fast
**Dark Fiber**: Physical fiber lease; costly; ISPs prefer selling lambdas
**Coarse Wavelength Division Multiplexing (CWDM)**: x16, cheaper, 120km, passive MUX, use mirrors to aggregate/separate signals (analog)
**Dense Wavelength Division Multiplexing (DWDM)**: x80, 10GB/s, 1000km, active MUX, more precise, uses software ("smarter"), wideband
**Connection-oriented**: Predefined path, packet IDs (ATM, Frame Relay)
**Connectionless**: No setup; full address in each packet (Ethernet, MPLS VPN)
**Circuit-Switched**: Requires signaling protocol to establish e2e circuit
**Packet-Switched**: Expects network to know how to forward packets

### PUBLIC WAN

**Site-to-Site VPN**: Fixed locations, devices locked to IP addr
**Remote Access VPN**: Changing locations, devices not locked to IP addr
**Software-Defined WAN (SDWAN)**: connects LANs across large distances using controlling software that works with a variety of networking hardware.

### MULTI PROTOCOL LABEL SWITCHING (MPLS)

– Advantage of eBGP at PE-CE: No mutual redistrib., same routing process

– Control plane (e.g. OSPF) → to learn labels
– iBGP to exchange NLRI (RD,RT,IPv4 Pref.,NextHop&VPN Lbl) between PE
– Traffic encrypted flowing over MPLS L3VPN backbone. (e.g. Bank)

**Label Switched Router (LSR)**: Forwards labeled packets
**Label Switched Path (LSP)**: Pre-determined path across MPLS network
**Unicast Reverse Path Forwarding (uRPF)**: checks source of each packet & verifies that source is in routing table
**imp-null**: networks are directly connected, no more label switching
**Virtual Private Wire Service (VPWS)**: Pseudowire (PW); No MAC learning at PE level; P2P L2 VPN, allows two sites to connect as if directly linked
**Virtual Private LAN Service (VPLS)**: Emulates LAN segment across MPLS backbone, provides multipoint L2 connectivity; MAC learning at the PE level

## ROUTER TYPES

**Customer Edge (CE)**: No knowledge of MPLS, no labels; connected to PE
**Provider Edge (PE)**: Connected to CE; runs iBGP and LDP; uses VRFs
**Provider or LSR (P)**: Inside MPLS VPN, no CE connection forwards labels

## DATABASES, PLANES

**Routing Information Base (RIB)**: Learned prefixes from routing protocols
**Forwarding IB (FIB)**: Built from RIB; only best routes for forwarding
**Label IB (LIB)**: All label mappings; 1 label per prefix
**Label Forwarding IB (LFIB)**: Built from LIB; used for actual forwarding decisions. (L)FIB only contains currently best LSP (decision: Routing Protocol)
**Control Plane**: Builds routing/label tables (RIB, LIB)
**Data Plane**: Forwards packets (FIB, LFIB); pushes/swaps/pops labels

## MPLS HEADER

4-byte header before IP. With MPLS VPN, there will be a stack of two headers
**Label (20b)**: actual MPLS label
**EXP (3b)**: QoS/CoS, Now called Traffic Class (TC)
**S-bit (1b)**: Bottom of label stack indicator, 1 = True = last label before IP header
**TTL (8b)**: time-to-live (equal to IP TTL)

## TTL

– Ingress PE decrements IP TTL field & copies IP TTL field into new MPLS TTL
– P routers decrements MPLS TTL
– Egress PE decrements MPLS TTL, pops final MPLS header, copies IP TTL
– If provider doesn't want to expose MPLS network: disable ***MPLS TTL propagation*** on PE, PE sets MPLS TTL=255, egress PE leaves original IP TTL unchanged ⇒ network appears as single router hop from TTL perspective

## LABEL DISTRIBUTION PROTOCOL (LDP) - CONTROL PLANE

– Distributes labels to neighbors, triggerd by new IP route in FIB
– ***Hello messages*** sent via UDP (Port 646) to `224.0.0.2` to discover neighbors
– TCP (646) connection used to exchange label bindings (prefix to local lbl)
– Routers advertise all local bindings after TCP session is up
– Label mapping used to build LIB → LFIB
– LDP router ID must be reachable (via routing table)
– Each router manages local labels independently
– Relies on underlying routing protocol for best route & loop prevention

## MPLS L3 DATA PLANE

**Outer label**: Transport label (LDP); identifies LSP between ingress/egress PE
**Inner label**: VPN label (MP-BGP); identifies customer VRF
**Push**: Ingress PE; classify and label packets
**Swap**: P router; replaces label, forwards based on new label
**Pop**: Egress PE removes label; sends original packet to CE
**Penultimate Hop Popping (PHP)**: penultimate router removes the outer MPLS label before forwarding to egress PE, default enabled, ISPs disable it

## VIRTUAL ROUTING AND FORWARDING (VRF) TABLES

– Virtual router inside a PE. Maintains isolated RIB + FIB and separate routing process per CE (VPN isolation)
– Exists per MPLS-aware PE router; one per attached customer

## ROUTE DISTINGUISHER (RD) VS. ROUTE TARGET (RT)

**Route Distinguisher (RD)**: ***Uniquely identifies*** VPN routes (per VRF) – allows same IP prefix to be used in different VPNs. Can be IPv4 or ASN. Forms ***VPNv4*** NLRI (**MP-BGP**): ***RD:IPv4 prefix*** (12 bytes: RD 8b, IP 4b)
**Route Target (RT)**: Controls **route import/export** between VRFs. Used as ***extended BGP community path attributes***. Typically in the form ***ASN:nn*** or ***IP:nn***, e.g., `65000:100`, `1:10`
**How RTs Work**: Route is tagged with an RT when advertised by BGP. Other VRFs import the route if the RT matches their import policy. Allows overlapping or shared connectivity between tenants (e.g., shared services).

## OVERLAY TECHNOLOGIES

***Overlay network*** (virtual) sits on top of the ***underlay network*** (physical).
**Generic Routing Encapsulation (GRE)**: Encapsulates data packets, unencrypted, P2P. Enables usage of unsupported protocols. Carries passenger protocol/original packet inside carrier/transport IP header routed by underlay

## SEGMENT ROUTING (SR)

**Source routing**: Sender defines full path using Segment (= Instruction) List
**Segment Identifier (SID)**: Each SID = 1 instruction (e.g., forward via ECMP, specific iface, or to a service)
**Segment List**: Ordered SID list carried in packet header; defines full route
**State in packet**: No per-flow state, intermediate nodes follow SID instructions
**No new control plane**: Uses existing protocols (OSPF, IS-IS, BGP) with extensions; no LDP or RSVP-TE needed
**Data plane**: Can operate over MPLS or IPv6
**Simple but powerful**: Enables TE, fast reroute, policy routing

### SEGMENT LIST OPERATIONS

**Push**: Insert SIDs into packet; set active SID (top of list)
**Continue**: Active SID not yet completed; keep processing it
**Next**: Current SID completed; activate next SID in list

### SEGMENT SIGNIFICANCE

**Global Segments**: Known and supported by all SR nodes in the domain, Installed in forwarding tables across the network (e.g. "Forward packet according to shortest path to Node1"). Defined in the SR Global Block (SRGB) and should be consistent across all nodes
**Local Segments**: Defined and installed only on originating node, Not forwarded by others, but must be understood network-wide (e.g. "Forward packet on interface to Node2"). Defined in the SR Local Block (SRLB) and are specific to the local SR node.

### CONTROL PLANE SEGMENT TYPES

**IGP Prefix Segment**: Global SID tied to IGP prefix (multi-hop); all nodes install forwarding entries. "steer traffic along ECMP-aware shortest path"
**IGP Node Segment**: Global SID for a specific node (shortest-path forwarding)
**IGP Anycast Segment**: Global SID for a group of nodes; traffic sent to nearest
**IGP Adjacency Segment**: Local SID; direct link to neighbor
**L2 Adjacency SID**: Local SID for Layer-2 segment (e.g., Ethernet link)
**Combining Segments**: End-to-end paths can mix IGP and BGP segments. Traffic to BGP Anycast → more ECMP in data centers

### SR-MPLS

– Reuses existing MPLS data plane – no hardware change needed
– Segments = MPLS labels; Segment List = label stack (top = active)
– Segments distributed via IGP/BGP; no LDP necessary (but possible)
– Supports both IPv4 and IPv6 networks
– PUSH=PUSH, CONTINUE=SWAP, NEXT=POP

### BENEFITS OF SEGMENT ROUTING

**Benefits**: Simplification (removes protocols, simple operations, admin and mgmt), enhanced Traffig eng. (Delay, Bandwidth, Packet Loss, TE metric, Controller, Source-Node), Seamless deployment, Robust
**Source Routing**: Balances distributed intel with centralized optimization
**Traffic Engineering (TE)**: Optimizes network performance by analyzing and controlling data flow to reduce congestion and improve QoS

## DRAWBACKS OF TRADITIONAL NETWORKS

**Control Plane**: LDP/RSVP-TE adds complexity
**Scalability**: Per-flow/path limits growth; LSP, signaling overhead increase fast

## ETHERNET VIRTUAL PRIVATE NETWORK (EVPN)

Overcome flood-and-learn limitations, doesn't rely on data plane learning, utilizes robust control plane MP-BGP, works with different encapsulation techniques (VXLAN, MPLS), excellent scalability, L2 and L3 Support.

## LAYER 2

L2 bridging across L3 networks
**BGP Control Plane**: Distributes MAC info (no flooding → efficiency)
**VXLAN Overlay**: Encapsulates L2 in L3 UDP (data plane)
**Multi-Tenancy**: via VNI segmentation
**Redundancy**: All-active multihoming, ECMP, fast convergence

### USE CASES

– Multi-tenant datacenter interconnects (DCI)
– Extending L2 over WAN between remote sites
– Scalable, segmented L2 fabrics

### AUTODISCOVERY VIA ROUTE REFLECTORS

– RR reflects EVPN routes to other PEs
– RR doesn't participate in EVPN or pseudowires
– RR needs only ***address-family l2vpn evpn***
– L2VPN RIB stores endpoint/VFI info for control plane
– ***BGP_UPDATE*** from spines contain ***ORIGINATOR_ID*** (origin leaf)

### HOST DETECTION

– Host connects to VTEP → MAC learned locally
– VTEP advertises MAC + L2VNI via BGP EVPN
– MAC learning follows normal Ethernet semantics

### INGRESS REPLICATION (IR)

Forwarding BUM traffic. When Multicast underlay network is not used, IR handles multi-destination traffic (ARP → unicast). Ingress device replicates BUM packets, sends them as separate unicast to egress devices.

### EARLY ARP TERMINATION (ARP SUPPRESSION)

– Avoids flooding ARP requests
– VTEP queries control plane for MAC/IP/VNI mapping
– If known → direct unicast (no broadcast)

***Silent Host Flow (Fallback)***

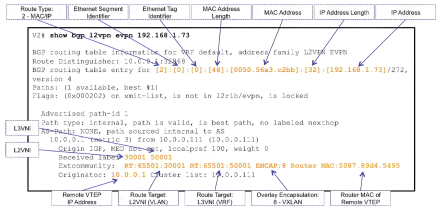
– If IP/MAC unknown → ARP sent via **ingress replication**
– Replicated ARP request goes to remote VTEPs
– Only correct host responds → update reflected to all VTEPs
– Future traffic uses updated BGP mapping

## LAYER 3

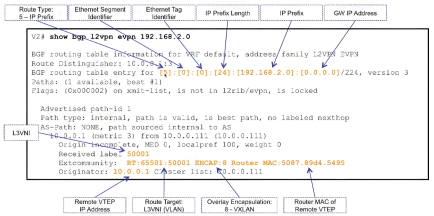
By adding L3 features, data routing efficiency + smooth connections improve

### EVPN ROUTE TYPES

(1) Ethernet Auto-Discovery Route (3) Inclusive Multicast Route (4) Ethernet Segment Route (6) Selective Multicast Ethernet Tag Route (7) IGMP Join Synch Route (8) IGMP Leave Synch Route
**Type 2 – Host Advertisement**: Advertises host MAC (mandatory), optionally IP, along with L2VNI and optionally L3VNI. Used for MAC learning, ARP suppression, and host mobility. Sent when host connects to VTEP.




**Type 5 – Subnet Advertisement**: Advertises IP prefix + prefix length with L3VNI. Used for inter-subnet routing. VTEP redistributes connected/static/dynamic IP routes. Additional attributes: L3VNI, extended communities.




### HOST DETECTION

– Host sends ARP/ND to local VTEP
– VTEP learns MAC/L2VNI and IP/L3VNI
– Info is advertised in EVPN (control plane)

### HOST DELETION & MOVE

**Host Deletion**: When a host detaches, its ARP (default: 1500s) and MAC entry (default: 1800s) time out on the VTEP. Upon aging, the VTEP withdraws the host's MAC/L2VNI and IP/L3VNI advertisements.
**Host Move**: When a host moves to a new VTEP, the new VTEP advertises updated reachability with a higher move sequence number. The old VTEP withdraws its entry, completing the migration.

### VIRTUAL ROUTING AND FORWARDING (VRF)

– Supports independent policies per tenant
– Key for scaling and multi-customer separation

### DISTRIBUTED ANYCAST GATEWAY (DAG)

– Same gateway IP+MAC on all VTEPs
– Enables local default gateway for hosts in MPLS network with IRB
– Supports mobility + optimal forwarding across BGP EVPN fabric

## INTEGRATED ROUTING AND BRIDGING (IRB)

– Enables inter-VLAN routing inside EVPN
– Avoids central gateway → no "traffic tromboning"

### SYMMETRIC IRB (L2 + L3)

– Routing/bridging on ingress + egress VTEPs
– Uses L3 Transit VNI (same in both directions), One L3 VNI per VRF (Tenant)
– Scales well; clean separation of MAC and IP
– Requires L3 connectivity between all source/destination VTEPs for Type 2 routing (Done by configuring Type 5 routing)

### ASYMMETRIC IRB (L2)

– Routing only on ingress, bridging on egress
– VXLAN uses destination VNI in both directions
– One L2 VNI per VLAN/Subnet
– Simple config, but requires all VLANs/VNIs on all VTEPs (scaling issues)

## MULTIPROTOCOL BGP (MP-BGP)

**Adress Family Identifier (AFI)**: Category of information being carried
**Subsequent AFI (SAFI)**: Narrows down the specific type within that category
**EVPN NLRI**: Uses MP-BGP with specific AFI/SAFI. Supports multiple route types and attributes, unsupported routes are dropped by BGP

– Enables protocol-based VTEP discovery and host reachability via control-plane learning. Reduces flooding by replacing data-plane learning
– Extends BGP with multiprotocol capabilities (AFI/SAFI)
– Uses ***MP_REACH_NLRI*** and ***MP_UNREACH_NLRI*** for route advertisement and withdrawal (advertises VPNv4 between PEs in MPLS)
– Uses the ***VPNv4*** NLRI to distinguish between duplicate IPv4 prefixes

### BGP CONTROL PLANE

– PEs learn MACs from local CEs (data plane)
– MACs advertised via BGP (control plane)
– Uses Route Distinguishers and MPLS labels
– Remote PEs update L2 RIB/FIB with MAC and next-hop info
– Enables seamless L2 across IP/MPLS backbone

## VIRTUAL EXTENSIBLE LOCAL AREA NETWORK (VXLAN)

Data plane technology which encapsulates Ethernet frames in UDP datagrams to tunnel layer 2 frames over a layer 3 network. The underlay network is unaware of the overlayt.

**Issues of L2**: STP, Max amount of VLANs (4094), Large MAC Address tables
**VXLAN**: Tunnels Ethernet (Layer 2) over IP using MAC-in-UDP encapsulation (Port 4789). For flexible and scalable network segmentation.
**VXLAN Network Identifier (VNID)**: 24-bit identifier (up to 16 million segments) that defines the VXLAN broadcast domain.
**Virtual Tunnel Endpoint (VTEP)**: Device (switch, router, or host) responsible for encapsulating/de-encapsulating VXLAN traffic.
**Network Virtual Interface (NVE)**: Logical interface on a VTEP used for VXLAN tunnel operations.

## FRAME FORMAT

– Original Ethernet frame → VXLAN Header → UDP → Outer IP Header
– The VXLAN header contains the 24-bit VNID and flags.
– Outer headers allow Layer 2 frames to traverse IP underlay networks.

## VIRTUAL NETWORK IDENTIFIER (VNI)

– 24-bit VXLAN Network Identifier uniquely defines VXLAN segments.
– Replaces traditional VLAN IDs (12-bit)
– Used by VTEPs to map traffic into corresponding Layer 2 domains.

## VXLAN TUNNEL ENDPOINT (VTEP)

Connects the overlay (VXLAN) and underlay (IP) networks. A VTEP can have multiple VNI interfaces, but they associate with the same VTEP IP interface.
**Software VTEP**: Located on hypervisors using virtual switches.
**Hardware VTEP**: Located on routers/switches with ASICs for performance.
**VTEP IP Interface**: Connects to the underlay network, handles encapsulation.
**VNI Interface**: Virtual interface per segment (like SVI); handles segregation of Layer 2 domains.

## MAC ADDRESS LEARNING

Each VTEP maintains a VXLAN mapping table linking destination MAC addresses to remote VTEP IPs.
**Data plane**: reactive, Flood and learn, MAC learned from incoming traffic, BUM via ingress replication or multicast underlay, No control-plane needed
**Control plane**: proactive. MP-BGP EVPN/MAC-to-VTEP advertised via BGP, No flooding needed for known MACs, ARP suppression, silent-host handling
**Flood and learn**: Host H1 sends ARP request, switches learn H1's MAC → ARP request is flooded to H2. → H2 responds; switches learn H2's MAC.
**Static VXLAN**: Manual MAC-to-VTEP mappings. Sends copy of BUM traffic to all participating VTEPs. Doesn't scale well; BUM traffic is inefficient.
**Multicast VXLAN**: VTEPs join multicast groups per VNI. Scales better, offloads BUM replication. 20+ VTEPs = too much traffic, doesn't scale well
**MP-BGP EVPN**: Modern solution using BGP as control plane. Dynamically learns MAC/IP info.

## CDN

**Origin Server**: Central content source (original files), usually in a datacenter
**Edge / CDN Server**: Also called ***Point of Presence (POP)***. Geographically distributed, caches content
**Embedded PoP**: Deployed directly inside ISP's local network
**DNS Infrastructure**: Directs users to optimal edge server (e.g. Geo-Routing)

## KEY BENEFITS

**Latency Reduction**: Nearby edge servers reduce round-trip time
**Availability**: Failover and redundancy in case of node failure
**Scalability**: Handles traffic spikes via load balancing
**Cost Optimization**: Reduces backend and transit load on origin
**DDoS Protection**: Edge servers absorb attacks → not all traffic on one server
**Global Load Reduction**: Less long-distance traffic across the Internet

## REQUEST ROUTING TECHNIQUES

– Decides which edge server should serve a client request
– Goal: Best performance (e.g. proximity, load, responsiveness)

### DNS-BASED GEO-ROUTING

– Each edge has a unique IP
– DNS server picks closest/optimal edge server based on:
  – Resolver IP location (not user! → can cause wrong choice)
  – GeoIP DBs (MaxMind, IP2Location), load, latency, business rules
– **EDNS(0) and Client Subnet Extension (ECS):**
  – Resolver includes part of client IP in DNS request (e.g. /24 subnet)
  – Authoritative DNS makes better decision based on actual client region
  – Improves accuracy without revealing full IP, increases caching complexity

### ANYCAST WITH BGP

– Same IP (e.g. `7.7.7.7`) advertised from multiple locations
– BGP routing decides which path is "best" (AS-path, local pref, etc.)
– No DNS logic or per-client decision – pure BGP convergence
– **Pros:** Fast failover, simple, no app logic needed
– **Cons:** Less control, BGP ≠ best latency, route flapping risk

## MANAGING TRAFFIC

**Anycast**: Modify BGP attributes/withdraw IP prefix. Not precise
**Software**: Shifting via Encapsulation tunnels (GRE)

## LOAD BALANCING

**Load balancing device**: Efficient, reliable, High cost, single point of failure
**ECMP**: ToR router balances traffic. Simple, cheap, Adding/removing server changes hashing pool, re-hashing breaks TCP connections fix: GRE tunnel

## HTTP CACHING & HEADERS

Caching is controlled via HTTP headers between clients, proxies, and servers
**Cache-Control**: Main directive (***no-cache***, ***no-store***, ***max-age***, ***must-revalidate***, etc.)
**Expires**: Absolute expiration time (older method, replaced by ***Cache-Control***)
**ETag**: Validator tag (version/hash), used with ***If-None-Match***
**Last-Modified**: Timestamp used with ***If-Modified-Since*** for revalidation
**Age**: Time (in seconds) since response was fetched from origin
**Validation**: Client uses ***ETag*** or ***Last-Modified***; server → 304 if unchanged

## QOS

**Internet is best effort**: No guarantees, no QoS; all traffic treated equally (net neutrality); simple, scalable, but no delivery/order assurance or prioritization
**Quality of Experience (QoE)**: Perceived service quality from user perspective
**Route Pinning**: Keeps flow on a fixed path to prevent oscillation (don't switch immediately to "better" path)
**Routing-based**: Path selection; QoS-aware routing; Choose best path
**Node-based**: Packet treatment; Congestion mgmt; queuing/scheduling/policing/shaping; "What happens to packets at each hop?"
**PAK-Priority**: System queue (routers) that prioritizes routing protocol packets
**Class models**: Realtime (VoIP > Video) > Control > Critical > Best effort

## NETWORK PERFORMANCE METRICS

**Bandwidth [Gbit/s]**: Maximum transfer capacity of a link (width of pipe)
**Throughput [Gbit/s]**: Successfully delivered data (water that reached dest)
**Latency / Delay [ms]**: Time for packets to travel src → dest (length of pipe)
  **End-to-End Delay**: Total time sender to receiver
  **One-Way Delay**: From first bit sent to last bit received
  **Components**: **Transmission delay** (time to push onto link), **Processing delay** (lookup, queuing), **Propagation delay** (physical travel time)
**Jitter [ms]**: Variation in delay between packets, caused by re-routing/queuing (Voip<30ms), Calc: no queue - queued delay. Doesn't impact TCP
**Packet Loss Ratio (PLR) [%]**: Dropped packets due to congestion or errors

## QUEUING ALGORITHMS

**First-In First-Out (FIFO)**: Basic, no prioritization
**Priority Queuing (PQ)**: Multiple queues, serve highest first; others may starve
**Round-Robin (RR)**: One packet per queue in turn (fair, but ignores priority)
**Weighted Fair Queuing (WFQ)**: RR with weights, *n* packets per queue
**Class-Based WFQ (CBWFQ)**: WFQ with user-defined classes (logical queues), queue limits, max bandwidth guaranteed or max % of bandwidth
**Low Latency Queuing (LLQ)**: Adds strict priority queue (priority class) to CBWFQ for delay-sensitive traffic (e.g. voice) (based on IP Precedence, DSCP, src, port, protocol…)

## QUEUE MANAGEMENT

**Tail Drop**: Drops packets when queue full; huge interruption of traffic → same as no connectivity
**TCP Global Sync**: Many TCP flows back off and restart simultaneously (AIMD) → link underutilization
**TCP Starvation**: TCP slows down after drops (congestion control), UDP doesn't → queues filled with UDP, TCP squeezed out
**Random Early Detection (RED)**: Random early drops before full queue to prevent global sync and TCP collapse. Dropped TCP segments cause TCP sessions to reduce their windows sizes
**Weighted RED (WRED)**: RED + DSCP/EXP-based drop logic, prioritizes higher-marked traffic, max thresh → all packets dropped

## POLICING VS. SHAPING

**Policing (Inbound mostly)**: Drops packets that exceed configured rate limits
**Shaping (Outbound)**: Buffers packets to smooth traffic bursts

## QOS MODELS

### BEST EFFORT

No guarantees, all traffic treated equally (follow Internet neutrality)

### INTEGRATED SERVICES (INTSERV)

– End-to-end QoS (guaranteed for specific apps)
– Per-flow resource reservation
– Precise but not scalable (uses RSVP)
– App informs network of traffic traits, requests service before sending data

### DIFFERENTIATED SERVICES (DIFFSERV)

– Scalable approach using marking, No hard guarantees, Cost-effective
– Divides traffic into classes based on business requirements
– Routers are configured with policy on how to prioritize classes

***Marking (DSCP)***

**L3 Marking**: ToS byte → DSCP (6 bits) + IP Precedence (3 bits) [overlapping]
  **DSCP**: (6-bit in IP header) marks packets for QoS; classifies traffic
**L2 Marking**: Dot1q header → 802.1p/CoS bits
**EXP**: (3-bit in MPLS label) serves same purpose within MPLS networks; often mapped from DSCP.

***Behavior (PHB)***

**Per-Hop behavior (PHB)**: Defines what routers actually do
  **Class Selector (CS)**: Compat with old IP Precedence, simple prio levels
  **Assured Forwarding (AF)**: Multiple classes + drop priorities
  **Expedited Forwarding (EF)**: Very high prio, e.g. VoIP

## MODULAR QOS CLI (MQC)

**Class Map**: Define traffic classes (e.g., match voice or video)
**Policy Map**: Define actions for each class (e.g., limit, shape, priority)
**Service Policy**: Apply policies to interfaces or directions (in/out)

## OTHER INFOS

**AD**: Inter-protocol choice (e.g., OSPF vs RIP) → lower wins.
**Cost/Metric**: Intra-protocol choice (e.g., OSPF path A vs B) → lower wins.
**Routing Preference Order (across protocols)**:
  – Most specific prefix
  – Lowest Administrative Distance
  – Static default route

| Protocol | Distance |
|---|---|
| Connected | 0 |
| Static (Interface) | 1 |
| Static (Next Hop) | 1 |
| BGP External | 20 |

**Administrative Distances**: (Smallest Administrative Distance wins)

| Protocol | Distance |
|---|---|
| EIGRP Internal | 90 |
| OSPF | 110 |
| ISIS | 115 |
| RIP v1/v2 | 120 |
| EIGRP External | 170 |
| BGP Internal | 200 |

## EVPN BGP ROUTING TABLE INFOS

* = Would not be there if it was L2 VNI BGP Routing Table
**Route Distinguisher**: `172.16.255.101:32777`
**Route Type**: 2
**MAC Address Length**: 48
**MAC Address**: 5254.00f8.29a8
***IP Address Length**: 32
***IP Address**: `10.10.0.100`
**L2 VNI**: 30010
***L3 VNI**: 50000
**Remote VTEP IP Address**: `172.16.254.101`
**L2 Route Target**: 1:10
***L3 Route Target**: 65000:50000

```
leaf-03# show bgp l2vpn evpn 10.10.0.100
BGP routing table information for VRF default, address family L2VPN
EVPN
Route Distinguisher: 172.16.255.101:32777
BGP routing table entry for
[2]:[0]:[48]:[5254.00f8.29a8]:[32]:[10.10.0.100]/272, version 19897
Paths: (1 available, best #1)
Flags: (0x000202) (high32 00000000) on xmit-list, is not in l2rib/evpn,
is not in HW
Advertised path-id 1
Path type: internal, path is valid, is best path, no labeled nexthop
Imported to 2 destination(s)
AS-Path: NONE, path sourced internal to AS
172.16.254.101 (metric 81) from 172.16.255.1 (172.16.255.1)
Origin IGP, MED not set, localpref 100, weight 0
Received label 30010 50000
Extcommunity: RT:1:10 RT:65000:50000 ENCAP:8 Router MAC:5254.00ca.69ae
Originator: 172.16.255.101 Cluster list: 172.16.255.1
```

## TODO'S

### OSPF
- passive interfaces
- flooding
- route summarization
- synchronizing the lsdb

### BGP
- comparison to IGP
- Multihop sessions
- NLRI
- Aggregate impact (TE)
- rework
  - Best path calculation
- Idle → Connect → Active → OpenSent → OpenConfirm → Established

### MULTICAST
- Link-local multicast (48)
- IGMP and the querier/Role in Dense Mode/Challenges

### EVPN
- IRB diagram

### QOS
- Types of traffic / QoS classification
- Queuing input/output buffer
- rest

***TODO:***
- github repos for IS-IS and OSPF labs
- why ibgp needed for ebgp
  - ip address otherwise might not be known in AS internally
- cdn lab (tags?)
- qos: why do we need it?
  - how does it work conceptually
- voice QoS sensitive to?
- spick lukas QoS+
- cisco config: bgp (slides VXLANEVPN 34), mpls, (ospf)

***TODO:***
notes 37+